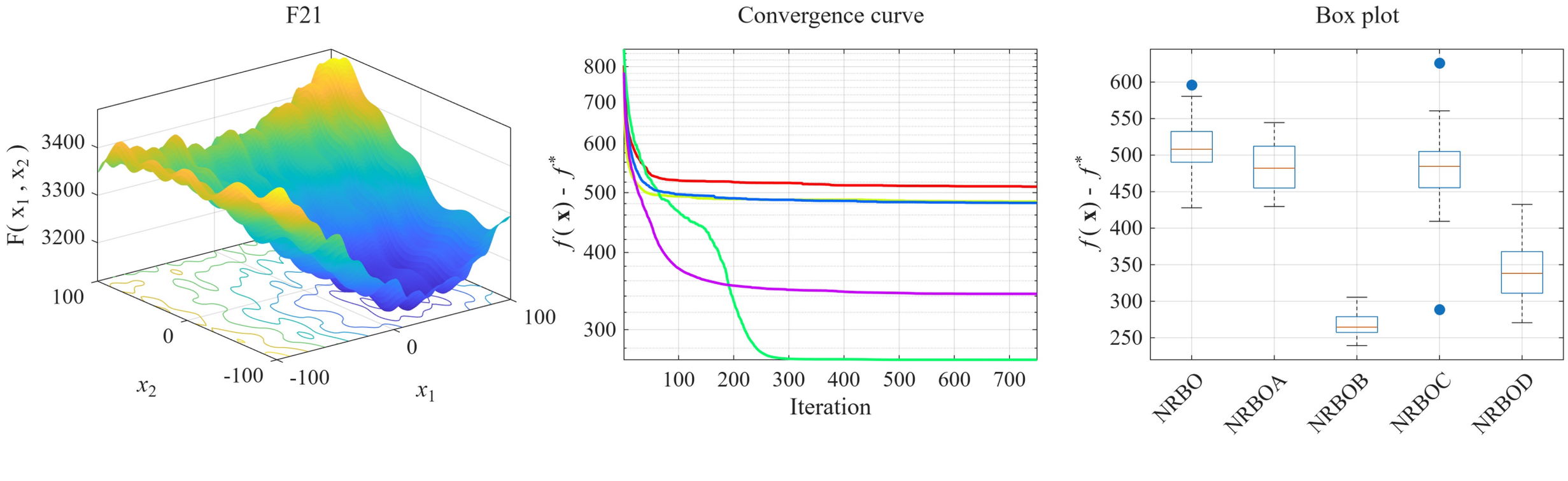

F21
F(x1 , x2)
3400
3300
3200
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
800
700
600
500
400
300
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
600
550
500
450
400
350
300
250
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD